Elements of the Paper	**%**
Style & structure ... which includes:	**15**
Uses the Junior Lab Latex Template	
Paper is 4 pages long, excluding title, abstract, acknowlegements and references. No Appendices.	
Paper consists of title, author list, abstract, introduction/motivation, theory/physics, detector description, calibration/data analysis, error analysis, results, discussion, conclusions, acknowledgements, references.	
Text is in prose. Paper does not consist of many bulleted lists. Paper is not in outline form.	
Paper does not suffer from spelling mistakes, grammar mistakes, or latex mistakes	
Figures are legible based on font size, choice of symbol, linetype, and color; Figures are readable in black &white	
Latex numbering is used on equations, figures, tables and citations	
Figures and tables are embedded in the text and have captions.	
All plots have axes labels with units, legends and titles.	
Variables and acronyms are defined throughout text	
All plots with overlayed fits include the fit form, resulting parameters, chi2 and dof in legible text on the plot	
Captions succintly describe the figures and tables.	
All relevant information appears in the main text (as opposed to in captions,long footnotes, etc).	
Overall clarity and professional tone.	
Introduction/Motivation and Theory/Physics ... which includes:	**5**
Introduction is not a long historical report	
The physics of and context for the measurement are correctly and succinctly explained	
The specific measurement goals are stated with clarity	
All relevant equations are quoted or derived, but lengthy derivations are avoided unless they are original and/or relevant	
Experimental Setup, Raw Data and Calibration ... which includes:	**20**
Sufficient information on the set-up is given such that someone outside of MIT Junior Lab could construct the experiment	
A labeled figure of the apparatus is provided.	
The main components of the apparatus are described in the text	

The detector physics for each major apparatus component is explained in the text

Procedure for acquiring data is explained

Example of raw data is shown as a figure in the paper. All raw data can be found in your log book (use notes in logbook to help us find it).

Data calibration is explained, with calibrated data shown as figure, if relevant

Statistical Analysis of Data ... which includes: **10**

Analysis procedure is explained step-by-step such that someone outside of MIT Junior Lab could repeat your method if given your raw and calibrated data.

The statistical uncertainty on the measurement is correctly calculated and reported.

Analysis of Systematic Uncertainty ... which includes: **15**

The list of systematic uncertainties is comprehensive and well-considered.

Sources of systematic uncertainties are listed in a table by priority. (If there are many, introduce catagories)

Numerical estimates for ALL of the systematic uncertainties are reported in the table. Associated discussion in the text explains how these uncertainties were quantified. Justification is provided for any uncertainties deemed negligible.

On plots, points are shown with statistical error while systematic uncertainties for each point are clearly displayed.

Fitting and presenting final results ... which includes: **10**

Parameters of fit results are reported. The chi2 and the dof for the fits are reported (separately--don't divide them!). The chi2 probabilities are also reported.

A discussion of the implication of the chi2 probability is included (eg "the probability indicates an acceptable fit" or "the probability indicates the systematic uncertainty may be underestimated", etc.)

Results are clearly stated with separate statistical and systematic uncertainties.

Calculational steps omitted in the paper for clarity are easily found in your logbook

Discussion of Results ... which includes: **10**

Physics implications of the result are discussed clearly and correctly.

Results are compared to accepted values. The deviation is quotes in sigmas for the total error (statistical and systematic combined)

Deviations of the result from the accepted values are discussed with respect to known systematic uncertainties, and improvements to the experiment with respect to these uncertainties are suggested.
Ideas for improving the experiment are included here, not the conclusion.

Title, Abstract, Conclusion, Acknowledgement and References ... which includes: **10**

Title is succcinct and encapsulates the goals of the study

Abstract and conclusion text succinctly reviews the salient points of the paper and restate the results, including deviation from expectation.

Include a statement in the acknowledgement listing out-of-class tools (writing clinic, Grammarly, chatGPT, etc.)

Citations appear for all research, ideas, data or information that are not your own.

References are complete and can be easily found with the information provided. References are from scientific sources (Wikipedia may only be used for figures illustrating the concepts)

Above and Beyond ... which includes: **5**

Original effort that goes beyond what is presented in the Junior Lab manual is rewarded! (Hard to get points)